Economic Analysis of Self-Help

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I. Research Motivation

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What is self-help?

- Rights are infringed
- No neutral third party to resolve the dispute
- Rely on own force or effort
- Without going to court or arbitration

Examples

- Pickpocket
- “Okupas”(Squatting)
- www.airhelp.com

I. Research Motivation

Questions …

- Why do so many people choose to use self-help to defend their rights?
- How do self-help relate to other tools in the rights redress system?
- How should the law respond to self-help?

Why important?

- In many areas of extreme inequality and institutional backwardness
- In developed industries and new rights
- Law and justice outside courts

I. Research Motivation

Self-Help in the Redress System: A Game Theory Model

Choices in the Real World: A Behavioral Experiment

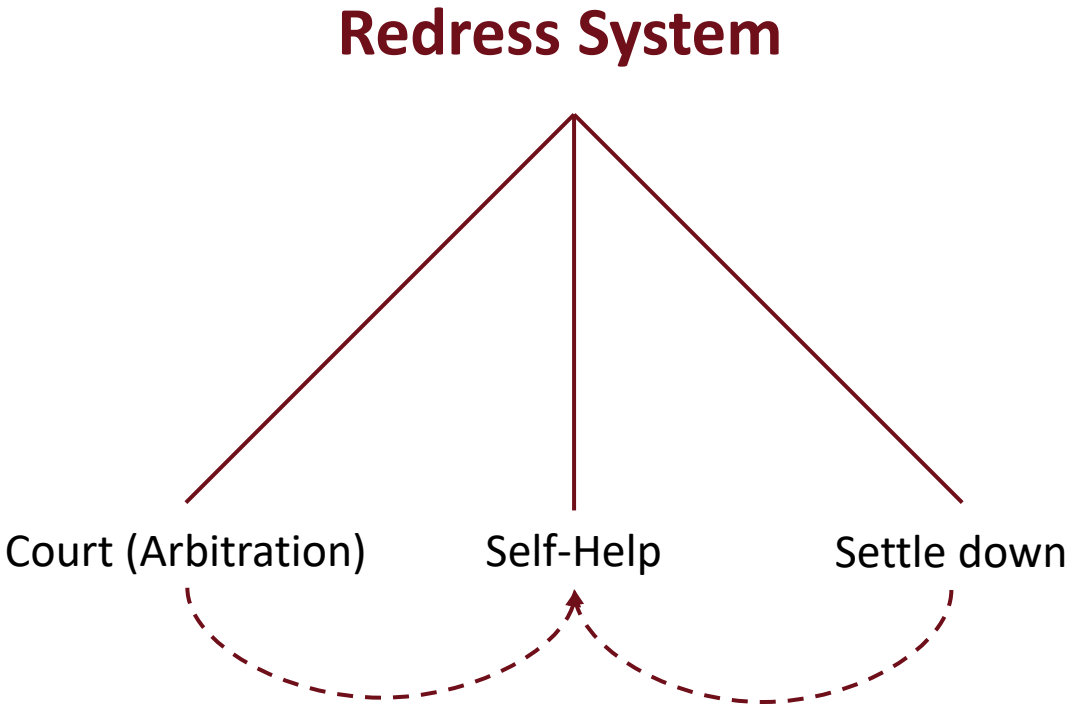
Legal Responses to Self-Help: Empirical Findings

II. Self-Help in the Redress System: A Game Theory Model

II. Self-Help in the Redress System

Systematic view of self-help

- “You think I'm trying to help myself? I was having to help myself!”



II. Self-Help in the Redress System

Court system

- **1:** the infringed party; **2:** the infringer
- **C_1:** cost and effort of the infringed party; **C_2:** cost and effort of the infringer
- **Payoffs in the very beginning:** (0, T)

How should the court decide?

- Give the property to the people who values it most!
- Let p represent the probability that C_1 win the game, then…
- $C_1 >> C_2 \Rightarrow p = 1$; $C_2 >> C_1 \Rightarrow p = 0$; $C_1 = C_2 \Rightarrow p = 0.5$
- $\frac{\partial p}{\partial C_1} > 0$ and $\frac{\partial^2 p}{\partial {C_1}^2} < 0$; $\frac{\partial p}{\partial C_2} > 0$ and $\frac{\partial^2 p}{\partial {C_2}^2} < 0$

$$\textbf{e.g. } p = \frac{C_1}{C_1 + C_2}$$

II. Self-Help in the Redress System

Court system (cont.)

- Rule: the people who puts more effort in sue wins the game; the lost pay both parties' cost.
- For the infringed party:

$$E_1 = \frac{C_1}{C_1 + C_2} \times T + \left(1 - \frac{C_1}{C_1 + C_2}\right) \times (-C_1 - C_2) = \boldsymbol{\frac{C_1}{C_1 + C_2} \times T - C_2}$$

- For the infringer:

$$E_2 = \frac{C_1}{C_1 + C_2} \times (-C_1 - C_2) + \left(1 - \frac{C_1}{C_1 + C_2}\right) \times T = \boldsymbol{\frac{C_1}{C_1 + C_2} \times T - C_1}$$

- **And they want to maximize their expectation!**

II. Self-Help in the Redress System

Court system (cont.)

- For the infringed party:

$$\frac{\partial E_1}{\partial C_1} = \cdots = \frac{T \times C_2}{(C_1 + C_2)^2} \geq 0$$

- For the infringer:

$$\frac{\partial E_2}{\partial C_2} = \cdots = \frac{T \times C_1}{(C_1 + C_2)^2} \geq 0$$

- **The optimal strategy is to keep increasing inputs until the maximum => Chicken Game!**

II. Self-Help in the Redress System

What about settlement?

- Coase Theory …
- “Who are you to talk to me about settlement?”
- No incentive

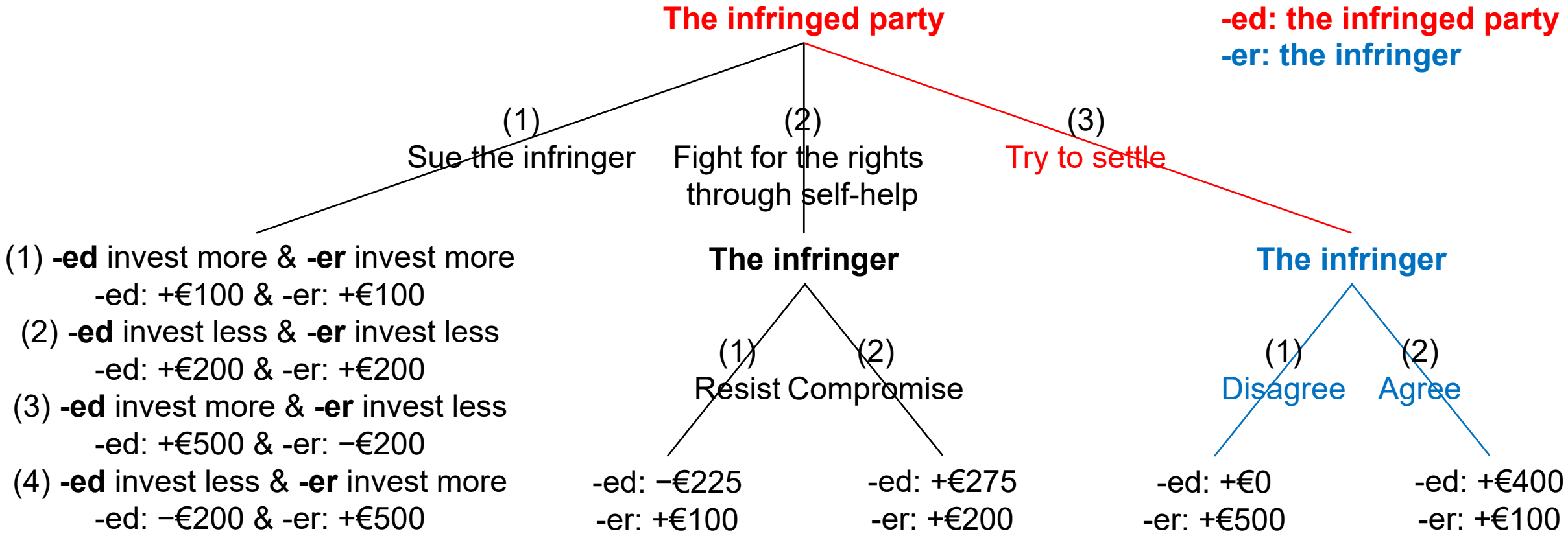
And self-help?

- Dynamic games
- Backward induction
- Threats harsh enough

=> Compromise!

III. Choices in the Real World: A Behavioral Experiment

III. Choices in the Real World



https://self-help-english.onrender.com/join/kizodini

IV. Legal Responses to Self-Help: Empirical Findings

IV. Legal Responses to Self-Help

Does more refined legislation on self-help leads to less inappropriate self-help?

- E.g., China Civil Code (2020)
- Difference in differences

Does reducing the expense of the justice system and increasing the speed of litigation help reduce inappropriate self-help?

- Online litigation reform
- Many empirical method

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Wish you a happy day!

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